

Mid Semester Examination 2025

Name of the Program: B.Tech CSE (AIML)

Semester: III

Name of the Course: Python Programming

Course Code: TCS346

Time: 1-1/2 Hour

Maximum Marks: 50

Note:

- i. All questions are compulsory
- ii. Answer any one sub-question among a, b from each main question
- iii. Total marks for each sub-question are 10 marks

Q1	(10 Marks)	COs
(a)	A company is designing an online feedback system for their homestay bookings. Guests often enter feedback in different formats (all lowercase, unnecessary spaces, repeated words, etc.). The raw feedback string is: <i>" moonlight homestay is amazing amazing place to relax "</i> Develop a menu-driven Python program using string methods to: i. Remove extra spaces at the beginning and end. ii. Replace duplicate word "amazing amazing" with a single "amazing". iii. Capitalize the first letter of each word properly. iv. Count the number of vowels in the cleaned string. v. Find the position of the word "relax". vi. Join the cleaned words with - as a separator and display the result.	CO1
OR		
(b)	At a metro ticket kiosk, fares depend on passenger type and distance: • Base fare: ₹5 per km • Discounts: Student 20%, Senior (age ≥ 60) 30% • Peak hours (07:00–10:00, 17:00–20:00) add a 10% surcharge after discount. Develop a Python program that: i. Takes name, age, passenger type ("student"/"senior"/"regular"), journey distance (float km), and travel time (HH:MM).	CO1

	ii. Computes final fare using the above rules with correct operator precedence. iii. Prints an invoice string (passenger, distance, time, category, final fare rounded to 2 decimals). Handle invalid inputs (e.g., negative distance, malformed time) with exceptions and proper messages.	
Q2	(10 Marks)	
(a)	Python provides both lists and tuples as sequence data types. i. Analyze the differences between lists and tuples with respect to mutability, memory usage, performance, and use-cases. ii. Illustrate your answer with at least two examples where a list is preferred and two examples where a tuple is more suitable. iii. Evaluate and justify in what type of real-life programming scenarios you would choose a tuple over a list.	CO3
	OR	
(b)	Python supports handling of both text files and binary files for persistent data storage. i. Compare and analyze text files and binary files in terms of storage format, readability, operations, advantages, and limitations. ii. Demonstrate how file opening modes differ ("r", "rb", "w", "wb", etc.) with suitable Python code examples. iii. Evaluate and justify with a real-world scenario when binary files are more appropriate than text files, and propose one application where text files are more suitable.	CO3
Q3	(10 Marks)	
(a)	Explain the concept of exception handling in Python. i. Define what an exception is and why exception handling is necessary. ii. Describe the role of try, except, else, and finally blocks with syntax. iii. Provide one simple example program to demonstrate how exceptions are caught and handled gracefully.	CO2

	OR	
(b)	Discuss the role of control statements in Python. - Define and describe the three main types: conditional statements (if, if-else, if-elif), looping statements (for, while), and jump statements (break, continue, pass). - Support your explanation with appropriate syntax and short examples for each type.	CO2
Q4	(10 Marks)	
(a)	An experiment logs pebble weights (grams) over a day: weights = [13.2, 12.8, 14.1, 13.9, 12.5, 15.0, 14.4, 12.9, 13.7, 14.2, 13.3, 15.1] Develop a menu-driven program to: - Show the first 3 and the last 3 readings using slicing. - Insert a new reading at a user-given position. - Find the index of a user-entered pebble weight. - Sort ascending and show min, max, mean (round to 2 decimals). - Save the cleaned list to weights_clean.txt (one value per line) and confirm count.	CO3
	OR	
(b)	Given raw log lines: [12-08-2025 10:45] Issue: "Deep Learning" by Ian Goodfellow, Roll: CSE034 [12-08-2025 10:52] Return: "Signals & Systems" by Oppenheim, Roll: ECE018 [12-08-2025 11:03] Issue: "Clean Code" by Robert C. Martin, Roll: CSE034 [12-08-2025 11:15] Fine: ₹50 paid by Roll: CSE034 Store the lines in a list and develop a Python program using re to: - Extract all actions (Issue/Return/Fine) and counts of each. - Extract every quoted book title into a list. - Validate and list all roll numbers matching $^[A-Z]\{3\}\d\{3\}\\$. - Compute total fine amount in the log (₹ prefix present). Display a neat summary.	CO1

Q5	(10 Marks)	
(a)	An e-commerce company receives thousands of user registrations daily. Many records have invalid emails, phone numbers, and postal codes. The company plans to use regular expressions for validation. - Explain why regular expressions are suitable for this problem compared to manual string checking. - Illustrate with examples how regex can be applied for email and phone number validation. - Discuss limitations of regex and suggest complementary methods to ensure data quality.	CO2
OR		
(b)	A university stores student attendance data in text files, while exam results are stored in binary files for faster processing. Recently, data corruption occurred due to mishandled file operations. - Explain the advantages and disadvantages of handling text vs binary files in Python. - Discuss how exceptions should be managed to ensure safe read/write operations. - Recommend best practices to maintain data integrity in such academic record systems.	CO